

Institute of Actuaries of India

Subject CB1-Business Finance

March 2022 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

<u>Solution 1:</u> C	[2 Marks]
<u>Solution 2:</u> B	[2 Marks]
<u>Solution 3:</u> A	[2 Marks]
<u>Solution 4:</u> C	[2 Marks]
<u>Solution 5:</u> D	[2 Marks]
<u>Solution 6:</u> C	[2 Marks]
<u>Solution 7:</u> B	[2 Marks]
<u>Solution 8:</u> A	[2 Marks]
<u>Solution 9:</u> D	[2 Marks]
<u>Solution 10:</u> B	[2 Marks]

Solution 11 : Project cost – Son's MBA = $-100,000 + 0 \cdot v + 0 \cdot v^2 + 50,000v^3 + 50,000v^4 + 50,000v^5$
 $= 2,762$

[2]

Project cost – Business Expansion = $-100,000 + (0.3 \cdot 0 + 0.7 \cdot 200,000) v$
 $= 27,273$

Where $v = 1/(1+10\%) = 0.9091$

[2]

Opportunity cost of Son's MBA = project cost of son's MBA – project cost of business
 $= 24,510 \text{ INR}$

[1]

[5 Marks]**Solution 12:**

- i) The assets of the company will be merged with the company. The negative net worth will be recorded as goodwill.
- ii) It should be mentioned in the notes to accounts, indicating the nature of the litigation, amount involved and the possible impact on company's future operations
- iii) Increase in building value on the asset side of the balance sheet. The increased value to be shown as revaluation reserve in the liability side.
- iv) Deducted as an expense in income statement based on accrual principle, just like any other interest due & shown as outstanding expenses in the liability side of the balance sheet.
- v) Capitalized in the balance sheet & amortized over a period of useful lifetime in income statement.

[1 Mark per correct point, max up to 5 Marks]

Solution 13:

Country	Profit Before tax (in lac)	Tax Rate	Tax amount in branch location (in lac)	Double tax relief	Additional tax in India, if any
India	15	20%	=15 * 20% = 3	NA/No	0
Dubai	5	05%	0.25	No	1
Sri Lanka	1	25%	0.25	Yes	0
Singapore	10	15%	1.5	Yes	0
Thailand	3	30%	0.9	No	0.6

[1 Mark per correct country, max up to 5 Marks]

Solution 14:

Technical Account Answer	Amount m Rs.
Earned Premium	10000
Investment Income Insurance	3000
Capital Loss on Reserves	-1500
Claims Incurred	-9000
Operating Expenses	-2000
Technical Acct Balance CF	500

[2.5]

Non-Technical Account	Amount m Rs.
Technical Acct Balance BF	500
Investment Income Equity	2500
Capital Gains on Free Reserves	1500
Profit Before Tax	4500
Tax on Profit	-1350
Profit After Tax	3150

[2.5]

[5 Marks]

Solution 15: Advantages:

- Easy to raise cheap capital for growth as the reach of the Company to various smaller shareholders' increases
- Easy to raise new debt from the market as lenders feel safer post compliance of stock exchange norms
- Gives existing shareholders an exit route as now the shares are exchange traded
- Make shares more marketable and easy to value as frequency trades happen on the exchange

Disadvantages:

- It increases compliance cost like auditor's fees as the Exchanges lays down compliance and disclosure norms
- It requires maintenance cost like annual Stock Exchange Fees to continue to be a part of the stock exchange, which is purely a cost of being present on the exchange

[0.5 marks for mentioning the point, 0.5 marks of explaining the point, max upto 5 marks]

Solution 16: $EBIT = \text{Sales} \times \text{Gross Profit Margin} - \text{Fixed Expenses} = 100,000 \times .3 - 15,000 = 15,000$

[0.5]

$\text{Return on Assets} = EBIT / \text{Total Assets} \Rightarrow \text{Total Assets} = 15,000 / .12 = 125,000$

[0.5]

$\text{Total Debt} = \text{Total Assets} - \text{Equity} = 125,000 - 50,000 = 75,000$

[0.5]

$\text{Mortgage interest} = (.1 - 0) \times EBIT = .1 \times 15,000 = 1,500$

$\text{Mortgage Debt Value} = \text{Mortgage Debt Interest} / \text{Interest \%} = 1,500 / .08 = 18,750$

$\text{Mezzanine Interest} = (.4 - .1) \times 15,000 = 4,500$

$\text{Mezzanine Debt Value} = 4,500 / .1 = 45,000$

[1.5]

$3\text{rd Priority Debt Value} = 75,000 - 18,750 - 45,000 = 11,250$

$3\text{rd Priority Debt Interest} = 11,250 \times .12 = \mathbf{1,350}$ [1 mark]

$\text{Interest Coverage Ratio} = 15,000 / (1,500 + 4,500 + \mathbf{1,350}) = \mathbf{2.04}$ times

[1]

[5 Marks]

Solution 17:

- i) The safety deposit money paid by both parties of a future contract to the clearing house as the contract requires future payment of money, is called Margin [1]
- ii) The 2 types of margin are: Initial margin, paid when the contract is first struck and Variation margin, paid daily to ensure control credit risk exposure of the clearing house
[0.5 marks for naming and 0.5 marks each for explaining each type, total 2 marks]

iii)

Day	Price of the bond	Your payment to clearing house
0	900	-100
1	1050	-50
2	1000	50
3	950	50
4	850	100
5	700	150
6	800	-100
7	900	-100
8	1050	-150
9	1000	50
10	1100	-100
11	1150	-50
12	1200	-50
13	1100	100
14	1050	50
15	950	100
Total		-50

[1 mark for showing calculation and 1 mark for correct total, total 2 Marks]

[5 Marks]

Solution 18:

Particulars		Amount Rs.
Profit After Tax	'252000-144000-48000	60,000
(+) Depreciation		60,000
(+) Increase in Payables		13,800
(-) Increase in Stock		-12,000
(-) Increase in Debtor		-9,000
Cashflow from Operating Activity		112,800
Purchase of Fixed Asset	'498000-280200+60000	-277,800
Cashflow from Investing Activity	s	-277,800
Share Capital Issued		120,000
Premium Generated		48,000
Cashflow from Financing Activity		168,000
Net Increase in Cash & Eq.		3,000
Cash at the beginning of the period		12,000
Cash at the end of the period		15,000

[5 Marks]**Solution 19:****i) Motives for growth:**

- i. Increase profitability & security
- ii. Increase motivation for employees

[0.5 marks per valid point, Max upto 1 Mark]

ii) Constraints in achieving growth may be:

- i. Difficulty in raising finance
- ii. Fear of fall in share price
- iii. Lack of managerial expertise
- iv. Limited time to prepare workforce
- v. Government restrictions

[0.5 mark per valid point, Max upto 2 Marks]

iii) ABC Ltd. can grow internally or externally.**[1]****Advantages of growing internally:**

- Retain control
- Avoid disruptive cultures/practices
- Avoid risk of losing integrity
- Avoid unnecessary government intervention

[0.5 mark for each valid point, Max upto 1.5 Marks]

Advantages of growing externally:

- Quicker form of growth
- Method of using spare cash
- Opportunity to share risk or financial burden of a project
- Opportunity to acquire resources (unique)

[0.5 mark for each valid point, Max upto 1.5 Marks]

[4]

iv) NPV is calculated as below:

Year	Profit in Cr.	Discount factor	Present Value
1	-6	$=1/(1+10\%)^1$ $=0.91$	$=-6*0.91$ $=(5.45)$
2	-2	0.83	(1.65)
3	1	0.75	0.75
4	3	0.68	2.05
5	10	0.62	6.21
Total	6		1.90

[0.5 marks for each year's calculation and 0.5 to total, max upto 3 Marks]

v) Considerations in an acquisition other than price:

- i. Any other resources (unique) being acquired
- ii. Any cost synergies involved
- iii. Additional bargaining power
- iv. Compatible values/managerial styles

[1 mark per valid point, Max upto 3 Marks]

vi) 4 Options to arrange for funds:

- i. Raise equity
- ii. Rights issue
- iii. Raise Debt
- iv. Use overdraft

[0.5 mark for each valid point, Max upto 2 Marks]

vii) Cost of capital = $10\% + 1.5 * 5\% = 17.5\%$ (risk free rate is mentioned in section d of the question)

[0.5 marks for calculation & total each, Total 1 Mark]

viii) Considerations to be made by Mr. G are as follows:

- i. Risk appetite of the share holders
- ii. Variability of returns expected in future
- iii. Share holders willingness to put additional capital in the business
- iv. Share holders willingness to reduce proportion of business that they hold

[1 Mark for each valid point, Total 4 Marks]

[20 Marks]

Solution 20:

i)

a. Inventory turnover ratio: [0.5 marks each]

- i. Definition of inventory turnover ratio
- ii. Need to understand the business model of the company. Is it just an assembly unit producing very little parts internally as compared to it's competitors?
- iii. Is the inventory management approach different to it's competitors? Is JIT being used?
- iv. Is there increased cost and time spent in managing inventory? How does this compare to the cost savings via lower inventory?
- v. Is the company losing out on discounts due to placing a large number of small orders?
- vi. Is the company consistently losing out on sales because of low inventory levels & unable to fulfill it's order book?

b. Creditor Turnover Ratio: [0.5 marks each]

- i. Definition of creditor turnover ratio

- ii. Longer creditor turnover ratio could indicate better negotiation ability for credit terms on the positive side.
 - iii. On the negative side, this could mean the company is unable to quickly pay its creditors on time
 - iv. What is the short term credit rating of the company?
 - v. Possibility of financing long term assets with short term credit
 - vi. Needs to be seen in the overall context of cash vs credit purchases
- c. Receivable turnover ratio: [0.5 marks each]
- i. Definition of receivable turnover ratio
 - ii. Is the company offering cash discounts for faster payments because it is unable to get short term financing at attractive costs?
 - iii. Is the company able to dictate terms of credit in the market place due to its reputation and product quality?
 - iv. What is the proportion of cash vs credit sales?
 - v. Does the company have a stringent policy on writing off delayed payments as bad debts quicker than its competitors?
- d. Measurement approach: [1 mark each]
- i. How are the ratios measured? Is it based on average flow during the year or the final value?
 - ii. Did the company have a massive clear out of inventory towards end of the period which depressed final stock?
 - iii. Is there a marked difference between the current period and previous period values?
- e. Financing, profitability & brand image: [1 marks each]
- i. It should have a significantly lower working capital position than its competitors
 - ii. Inventory & debtors are non-interest earning blockage of funds, while suppliers are an interest free source of funds.
 - iii. Cost of short term financing should be lower
 - iv. Overall, this should result in higher profit margins than its competitors
 - v. This is provided cash management and inventory management expenses are well managed
 - vi. Brand image of the company is likely to be better than its competitors due to its perceived efficiency levels relative to competition.
 - vii. However, in the event that the company is deliberately paying off suppliers late, the reputation of the company would be poor in the market place.
 - viii. Possibility of poor current/quick ratios indicating liquidity issues with long term questions on sustainability

[15]

ii)

- Maybe dismissed or undervalued if produced solely by finance team without involving operations
- Potential to lead to cost reduction & not value creation
- Involves many subjective forecasts & opinions
- Expensive to produce & monitor
- Requires a huge level of skilled personnel who understand the operations
- May reduce performance and encourage incremental thinking on the part of management
- Can lead to motivation issues among staff
- Slack budget reduce efforts, unduly stretched budgets discourage staff from trying

- Budget holders may be reluctant to exceed the budget in fear of next year benchmarks being higher
- Can reduce flexibility & act as barrier to change with potentially missed opportunities

[0.5 marks per valid point, max 5 marks]

[20 Marks]
